LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 5 – Solutions

- *Pretty straightforward, I believe!* You must complete this take-home quiz in **one continuous 30-minute attempt**. No breaks, no pauses, just you, your brain, your calculator and your A4 paper.
- You are strictly not allowed to seek help from **any external source**. This includes textbooks, online material, solution manuals, discussion forums, and all forms of AI/LLMs).
- You must **not discuss** the quiz with your peers or anyone else before, during, or after your attempt. *Academic **integrity*** applies everywhere!
- At the top of your first page, clearly write the following:
  - **Start Time:** _____
  - **End Time:** _____
  - **Venue of Attempt:** _____ (e.g., home, library, hostel room)
- Handover the quiz during the next class.
- If you are ready, please proceed to the next page.

Question 1 (3 marks)

Consider Soft SVM objective:

$$\min \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

subject to

$$y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Given data points $x_1 = (0,0)$, $y_1 = +1$, and $x_2 = (2,0)$, $y_2 = -1$.

(a) Show that no perfectly separating linear classifier exists with margin 1.

(b) Compute minimal slack ξ_1 and ξ_2 for classifier $w = (1,0)$, $b = -0.5$.

(c) For $C = 10$ compute the cost.

Solution: **(a)** For perfect separation with margin 1, we must have

$$y_i(w^\top x_i + b) \geq 1.$$

For $x_1 = (0,0)$ and $y_1 = +1$:

$$w^\top x_1 + b = b \geq 1.$$

For $x_2 = (2,0)$ and $y_2 = -1$:

$$-1 \cdot (2w_1 + b) \geq 1 \quad \Rightarrow \quad 2w_1 + b \leq -1.$$

Combining:

$$b \geq 1 \quad \text{and} \quad b \leq -1 - 2w_1.$$

These inequalities cannot hold simultaneously for any w_1, and therefore we can say that No perfect margin-1 separator exists.

(b) Given $w = (1,0)$ and $b = -0.5$:

For $x_1 = (0,0)$, $y_1 = +1$:

$$y_1(w^\top x_1 + b) = 1(-0.5) = -0.5.$$

Slack:

$$\xi_1 = 1 - (-0.5) = 1.5.$$

For $x_2 = (2,0)$, $y_2 = -1$:

$$w^\top x_2 + b = 2 - 0.5 = 1.5,$$
$$y_2(w^\top x_2 + b) = -1(1.5) = -1.5.$$

Slack:

$$\xi_2 = 1 - (-1.5) = 2.5.$$

(c) Soft SVM cost:

$$\frac{1}{2}\|w\|^2 + C(\xi_1 + \xi_2) = \frac{1}{2}(1^2) + 10(1.5 + 2.5).$$
$$= 0.5 + 10(4) = 0.5 + 40 = 40.5.$$

Question 2 (3 marks)

Consider the polynomial kernel of degree 2 defined as

$$K(x,z) = (1 + x^\top z)^2,$$

for input vectors $x, z \in \mathbb{R}^2$.

(a) Derive an explicit feature mapping $\phi(x)$ such that

$$K(x,z) = \phi(x)^\top \phi(z).$$

(b) For $x = (1, 2)^\top$ and $z = (2, 1)^\top$, compute $K(x, z)$ directly from the kernel definition.

(c) Using your mapping $\phi(\cdot)$ from part (a), compute $\phi(x)^\top \phi(z)$ for the same x and z, and verify that it matches the result from part (b).

Solution: (a)

Let $x = (x_1, x_2)^\top$ and $z = (z_1, z_2)^\top$. Then

$$K(x, z) = (1 + x^\top z)^2 = \big(1 + x_1 z_1 + x_2 z_2\big)^2.$$

Expanding the square,

$$\begin{aligned} K(x, z) &= 1 + 2(x_1 z_1 + x_2 z_2) + (x_1 z_1 + x_2 z_2)^2 \\ &= 1 + 2x_1 z_1 + 2x_2 z_2 + x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2. \end{aligned}$$

We can rewrite this in an inner-product form by grouping symmetric terms. Define the feature map

$$\phi(x) = \begin{bmatrix} 1 \\ \sqrt{2}\, x_1 \\ \sqrt{2}\, x_2 \\ x_1^2 \\ \sqrt{2}\, x_1 x_2 \\ x_2^2 \end{bmatrix}, \qquad \phi(z) = \begin{bmatrix} 1 \\ \sqrt{2}\, z_1 \\ \sqrt{2}\, z_2 \\ z_1^2 \\ \sqrt{2}\, z_1 z_2 \\ z_2^2 \end{bmatrix}.$$

(b)

We first compute the inner product:

$$x^\top z = 1 \cdot 2 + 2 \cdot 1 = 2 + 2 = 4.$$

Therefore,

$$K(x, z) = (1 + x^\top z)^2 = (1 + 4)^2 = 5^2 = 25.$$

(c)

Compute the inner product:

$$\begin{aligned} \phi(x)^\top \phi(z) &= 1 \cdot 1 + (\sqrt{2})(2\sqrt{2}) + (2\sqrt{2})(\sqrt{2}) + 1 \cdot 4 + (2\sqrt{2})(2\sqrt{2}) + 4 \cdot 1 \\ &= 1 + 4 + 4 + 4 + 8 + 4 \\ &= 25. \end{aligned}$$

Thus,

$$\phi(x)^\top \phi(z) = 25 = K(x, z).$$

Question 3 (4 marks)

Multiple options can be correct.

MCQ1: Hard SVM fails when:

(a) Data is normalized

(b) The margin becomes too large

(c) Data is not linearly separable due to noise or outliers

(d) Number of features is small

Solution: (c) Hard SVM is infeasible when data is not perfectly separable.

MCQ2: Gaussian (RBF) kernel corresponds to mapping into:

(a) 2D space

(b) Finite polynomial space

(c) Infinite-dimensional space

(d) 1D space only

Solution: (c) RBF kernel implicitly maps data into an infinite-dimensional feature space.

MCQ3: The decision function for SVM is:

(a) $y = w + x$

(b) $y = \text{sign}(w^\top x + b)$

(c) $y = x^\top x$

(d) $y = \text{hinge}(x)$

Solution: (b) SVM predicts using the sign of the affine score $w^\top x + b$.

MCQ4: Which of the following interpretations of the SVM weight vector **w** is correct?

(a) Magnitude of **w** controls number of support vectors

(b) **w** determines bias adjustment

(c) **w** determines the orientation (normal vector) of the separating hyperplane

(d) magnitude of **w** determines the classification margin

Solution: (c) and (d)